

SIMATS ENGINEERING

ECA09 - DIGITAL SIGNAL PROCESSING LAB

Experiment 02.2

Design and Performance Evaluation of a Chebyshev Type-I IIR High-Pass Filter.

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Aim

To design and analyze a Chebyshev Type-I IIR high-pass filter for the specified passband and stopband requirements and verify stability.

Procedure / Calculations

Specify the passband and stopband edges, calculate the minimum order with `cheb1ord()`, design the high-pass filter using `cheby1()`, and evaluate frequency, impulse, group-delay, and pole-zero plots.

MATLAB Program

```
clc;
clear;
close all;

% Specifications
Rp = 1;      % Passband Ripple (dB)
Rs = 40;     % Stopband Attenuation (dB)
Wp = 0.5;    % Passband Edge
Ws = 0.35;   % Stopband Edge
// NO:2 Chebyshev IIR Filter
// TE: 2.2. Design and Performance Evaluation of a Chebyshev Type-I IIR High-Pass Filter

% Find Filter Order
[n,Wn] = cheb1ord(Wp,Ws,Rp,Rs);

% Design High-Pass Filter
[b,a] = cheby1(n,Rp,Wn,'high');

% Frequency Response
figure;
freqz(b,a);
title('Frequency Response');

% Impulse Response
figure;
impz(b,a);
title('Impulse Response');

% Group Delay
figure;
grpdelay(b,a);
title('Group Delay');

% Pole-Zero Plot
figure;
zplane(b,a);
title('Pole-Zero Plot');

fprintf('Filter Order = %d\n',n);
fprintf('Cutoff Frequency = %.4f\n',Wn);
```

Output / Analysis

The output/analysis pages below are reproduced from the uploaded official ECA09 MATLAB lab manual for this experiment.

```
fprintf('Cutoff Frequency = %.4f\n',Wn);
```

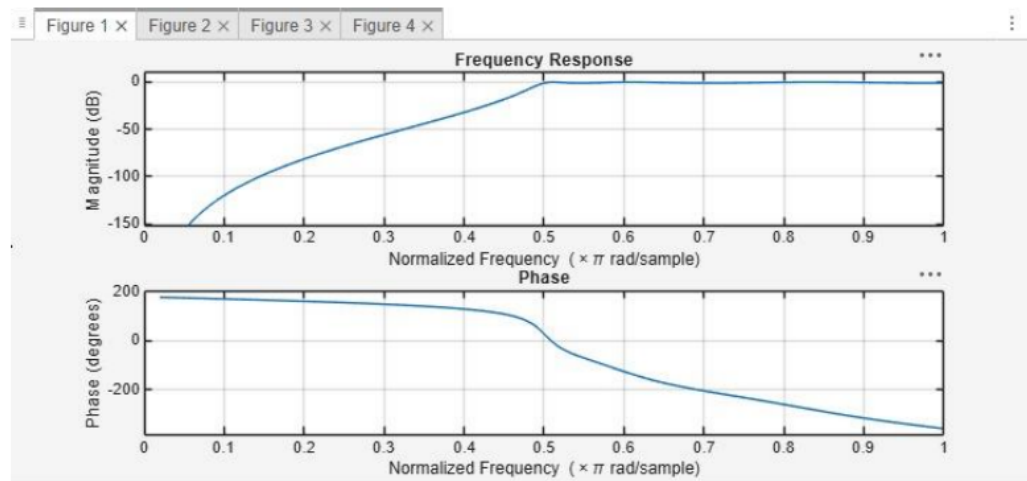
Output::

Objective-1:

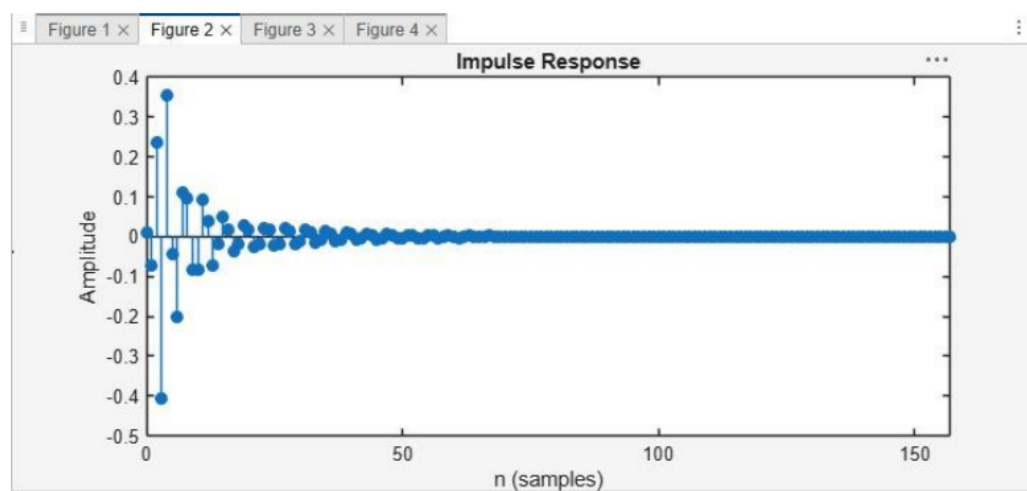
Filter Order = 6

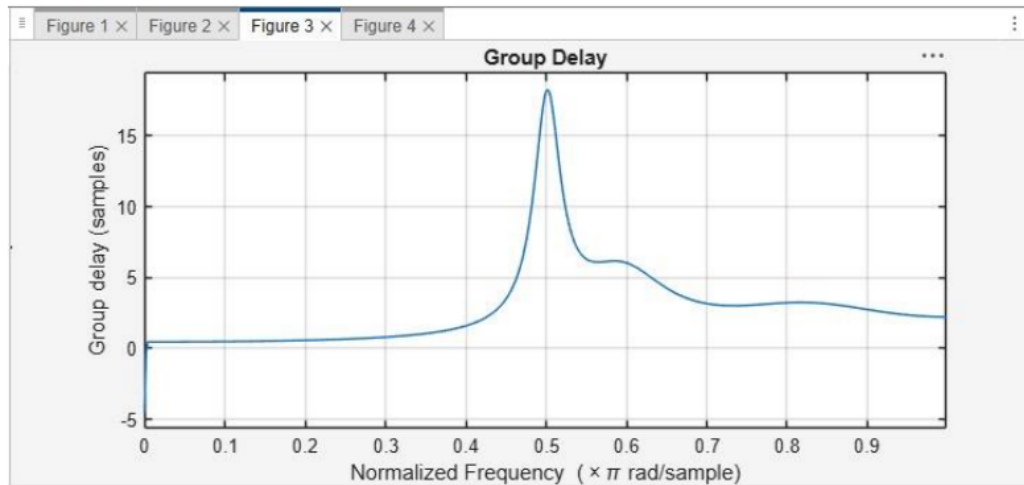
Cutoff Frequency = 0.5000

Objective 2:

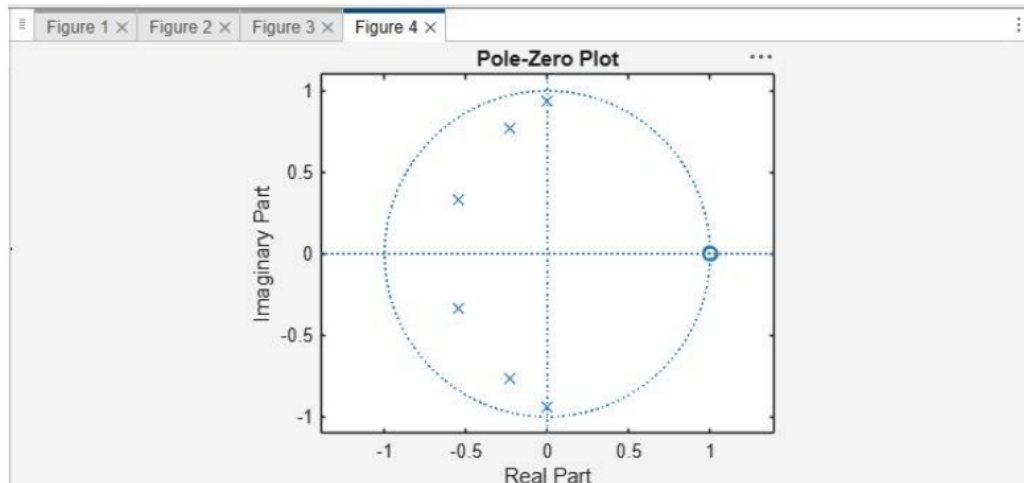


Objective 3:





Objective 4:



Result:

1. The Chebyshev Type-I IIR High-pass filter was successfully designed, and its magnitude response, phase response, impulse response, and group delay were analyzed.
2. The impulse response and group delay characteristics were obtained and found to satisfy the given specifications.
3. The pole-zero plot confirmed that all poles lie inside the unit circle, verifying the stability and proper performance of the filter.
4. The minimum filter order and cutoff frequency were successfully determined and found to satisfy the given design specifications.

Practical Question 1:

Design a digital **Chebyshev Type-I IIR High-Pass Filter** using MATLAB to meet the following specifications: passband ripple of **1 dB**, minimum stopband attenuation of **40 dB**, passband edge frequency of **2000 Hz**, stopband edge frequency of **1000 Hz**, and sampling frequency of **8000 Hz**.

1. Determine the minimum filter order and cutoff frequency required to satisfy the given specifications.
2. Obtain and analyze the magnitude response and phase response of the designed filter.
3. Obtain and analyze the impulse response and group delay characteristics of the designed filter.
4. Generate the pole-zero plot and verify the stability and performance characteristics of the filter.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Practical Question 2

Design a digital **Chebyshev Type-I IIR High-Pass Filter** using MATLAB to meet the following specifications:

- Passband Ripple = **0.5 dB**
- Minimum Stopband Attenuation = **45 dB**
- Passband Edge Frequency = **3000 Hz**
- Stopband Edge Frequency = **2000 Hz**
- Sampling Frequency = **10000 Hz**

Tasks

1. Determine the minimum filter order and cutoff frequency required to satisfy the given specifications.
2. Obtain and analyze the magnitude response and phase response of the designed filter.
3. Obtain and analyze the impulse response and group delay characteristics of the designed filter.
4. Generate the pole-zero plot and verify the stability and performance characteristics of the filter.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Result

The Chebyshev Type-I high-pass filter was successfully designed and analyzed. The reported cutoff/order and pole-zero plot demonstrate the intended high-pass behavior and stability.

Student Details

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